



Bangladesh University

Department of CSE
Final Examination, Fall Semester 2024

Course Code: CSE-2105
Course Title: Discrete Mathematics
Course Teacher: Akash Kumar Pal
Batch: 67 (A & B) (DAY)

Time: 2.00 hours

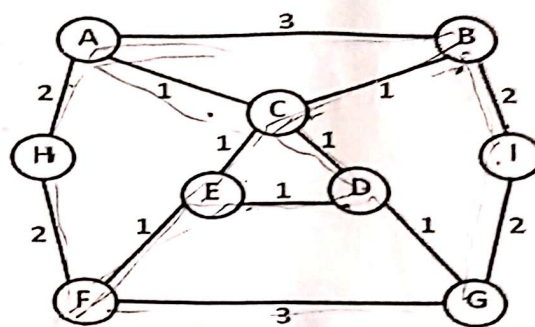
Total Marks: $5 \times 10 = 50$

Carefully read and follow all of the instructions given below.

- Carefully fill out the front page of the answer script with all the requested information.
- Read the questions attentively and answer accordingly.
- It is completely forbidden to employ any unfair methods during the exam.
- Make sure to keep the exam room noise-free, and BEST OF LUCK.

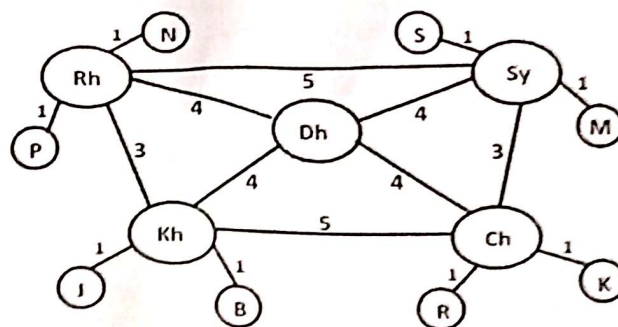
Answer any five questions from below.

- "Graph is a nonlinear data structure" justify this statement with example. 2
 - State real life applications of graph & what advantages/benefits we can gain from using graph? 3
 - Draw 4 types of graph & state their key characteristics. State the basic properties by which we define characterization of a graph. 5



Graph T

- Calculate the distance between Vertex B to Vertex F from Graph T? [N.B.: Show all possible path]. 3
- Identify the Eccentricity value of all Vertex of Graph T. 5
- Identify the Radius, Diameter, Central Point & Center of Graph T [N.B.: Utilize your findings from 2.(b)] 2



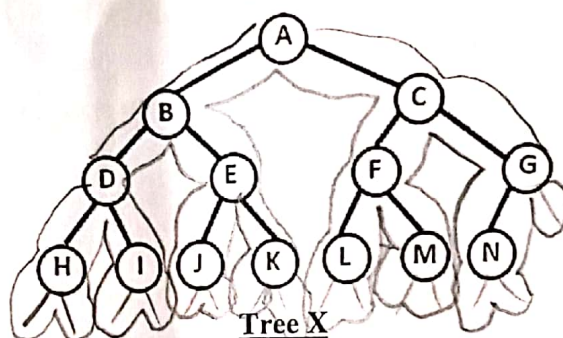
Graph BD

- What are the benefits that we can gain through graph representation techniques? Represent the given Graph BD through Adjacency Matrix [NB: All the edges are bidirectional]. 5

- (b). The given Graph **BD** represents the **Map of Bangladesh** where the **Dh** represents the **capital Dhaka**. The govt. wants to establish 6G infrastructure **throughout** the country **starting from the capital**. The govt. aims to reach the **divisions (Rh, Kh, Ch, Sy)** at first which are **directly connected to the capital** & then the **districts**. Select an appropriate method & demonstrate by which the govt. can reach all its destination. 5

4. (a). Define tree & state the **Depth & Level** of **Root** node in a tree. 2
 (b). State the **differences** between a **General tree** & a **Binary tree** with example. 3
 (c). Define & draw a **complete binary tree** & a **full binary tree**. Show that the below given expression can be represented through a **binary expression tree**: 5

$$[(I * J) / (K + L)] * (X + Y * Z)$$



acrylide

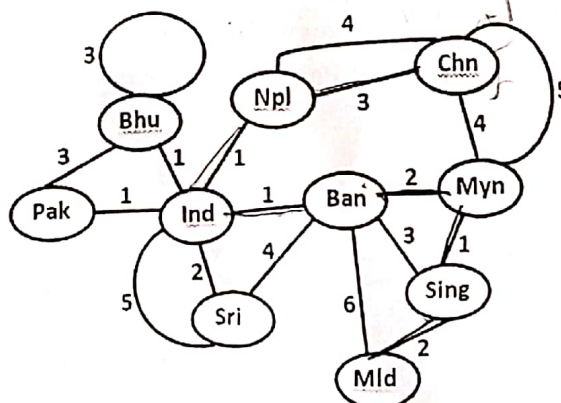
- (a). From the above given binary tree X identify the **leaves & siblings**. 2
 (b). **Traverse all the nodes** of the above given binary tree X in **Preorder, Postorder & Inorder** tree traversing technique. 5

(c). 3

Preorder	O	P	N	A	R	U	T	@	C	L	E	V	I
Inorder	A	N	R	P	T	U	@	O	E	L	V	C	I

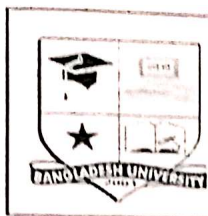
Given the data (**Preorder & Inorder**) provided above, construct the tree.

6. (a). Prove that a **complete undirected** graph can have maximum n^{n-2} number of spanning trees, where n represents the number of **nodes**. 2
 (b). Define **spanning tree & MST**. Is it possible construct a spanning tree from a graph that is **disconnected**? Justify your answer. 3



Graph Asia

- (c). **Dr. Yunus**, the chief advisor of Bangladesh (**Ban**) in his **Asia Peace Tour** wants to visit all the major **South East Asia countries** to strengthen the diplomatic bonds with Bangladesh. He has a **tight schedule**, so he wants this tour to be **time efficient and precisely scheduled**. Now provide a solution/ tour plan so that **Dr. Yunus** can conduct his **Asia Peace Tour** successfully. 5



Bangladesh University

Final Examination, Fall-2024

Course Code: CSE-2101

Course Title: Programming Language Java

Batch: 67th (DAY)

DEPT. OF

CSE

Total Marks: 50

Full Time: 2 hours

Instructions: 1. Answer any 5 questions out of the 7 questions.

2. Read the questions carefully and answer accordingly.

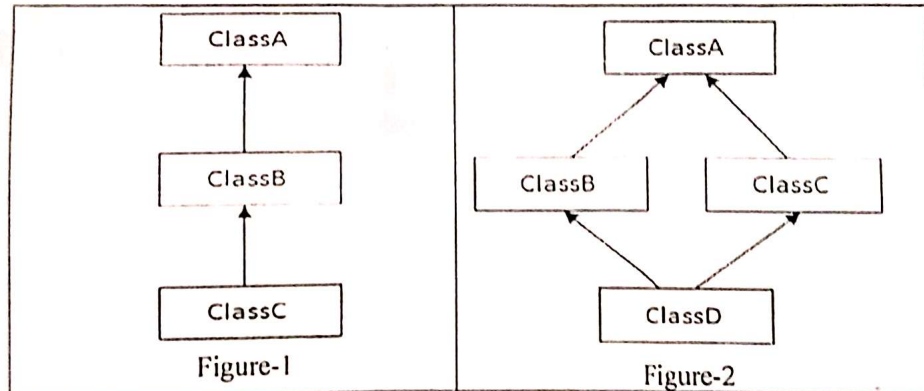
3. If you have problem in understanding any question, ask your teacher; not your classmate.

4. To be honest and confident with look and attitude.

5. Instructor Name: Pabon Shaha

1. a). Define the term OOP. 2
b). Object-oriented programming has several advantages over procedural programming. Justify the statement. 4
c). Mention some characteristics of an Object-Oriented Programming Language 2
d). Define Constructor. Write down the rules of Constructor 2
2. a). Write a java program to demonstrate the working of a banking-system. 4
✓ where we deposit and withdraw amount from our account.
✓ Creating an Account class which has deposit() and withdraw() methods.
b). What is the necessity to use method in java? 4
c). Can we overload java main() method? 2

3.



- a). What do you mean by inheritance? 2
b). Write a java code to implement the figure-1 scenario. 4
c). Describe the scenario which is mention in figure-2. 4
4. A program performs division where the divisor is zero, causing an **ArithmeticException**, and tries to access an invalid array index, triggering an **ArrayIndexOutOfBoundsException**. Both exceptions are caught using separate catch blocks to prevent the program from crashing. The program then displays appropriate error messages and continues execution.
- a). Define the term exception handling in Java. 2
b). How can you handle the exceptions mentioned in the scenario using code? 6
c). Mention the best practices for exception handling. 2

	Bangladesh University Final Examination, Fall-2024 Course Code: CSE-2101 Course Title: Programming Language Java Batch: 67 th (DAY)	DEPT. OF CSE
--	---	-----------------------------------

Total Marks: 50

Full Time: 2 hours

5. a). Define the term method overloading and method overriding
 b). Print the output of the following code-

3
4

<pre> class Bank { int getRateOfInterest(){return 0;} } //Creating child classes. class SBI extends Bank { int getRateOfInterest(){return 8;} } class ICICI extends Bank { int getRateOfInterest(){return 7;} } class AXIS extends Bank { int getRateOfInterest(){return 9;} } </pre>	<pre> //Test class to create objects and call the methods class Test2 { public static void main(String args[]){ SBI s=new SBI(); ICICI i=new ICICI(); AXIS a=new AXIS(); System.out.println("SBI Rate of Interest: "+ s.getRateOfInterest()); System.out.println("ICICI Rate of Interest: "+ i.getRateOfInterest()); System.out.println("AXIS Rate of Interest: "+ a.getRateOfInterest()); } } </pre>
--	---

- c). Explain different ways to overload the method.

3

6. a). Multiple inheritance is not supported through class in java, but it is possible by an interface, explain it.
 b). Correct and complete the code according to the statement. Print the output of the code after complete your task.

4
4

<pre> class Animal { void eat(){System.out.println("eating...");} } // here might be any correction class Dog implements Animal { void bark(){System.out.println("barking...");} } // Write a Babydog class which extend the Dog class and create a weep() method which print barking. </pre>	<pre> class TestInheritance3 { public static void main(String args[]){ BabyDog d=new BabyDog(); d.weep(); d.bark(); d.eat(); } } </pre>
---	---

- c). Why we use inheritance in Java?

2

7. a). Define the term abstraction. Write down the way to achieve abstraction.
 b). Discuss the abstract class with proper code.
 c). Explain the use of Java Interface.

3
4
3



Bangladesh University
Department of Computer Science & Engineering
Final Examination, Fall 2024
Course Code: CSE-2103, Course Title: Data Structure
Total marks: 50 Time: 2 hours

[Answer any *five* questions out of *six*.]

1. a) Define a queue and describe its characteristics in data structures. 2+8
b) Write a C++ program to implement the INSERT and DELETE operations of a queue in action.
2. a) What is a stack? Explain how it functions in data management. 2+8
b) Write a C++ code for stack POP and PUSH operation.
3. a) What is a tree in data structures? Write down the rules for binary tree construction. 2+8
b) Create a C++ program to build and display a tree structure.
4. a) Draw a binary search tree with the given values. 2+8
50, 45, 40, 30, 56, 20, 70, 60
b) Write the array updates with front and rear values for the following queue operation, where queue maximum size is 5.
 - i. A, B, C, D insert
 - ii. A delete
 - iii. E, F insert
 - iv. B, C delete
 - v. G insert
5. a) Consider the algebraic expression $E = [a + (b - c)] * [(d - e) / (f + g - h)]$. Draw the binary tree for the expression. 4+6
b) Write down the node representation and array representation for the tree.
6. a) What are the basic differences between tree and graph? 2+5
b) Draw a graph with five vertices and seven edges. Exactly one of the edges should be a loop, and do not have any multiple edges. Write down the degrees of each vertex. +3
c) Draw the directed graph that corresponds to this adjacency matrix:

	A	B	C	D
A	1	0	1	0
B	1	0	0	1
C	0	0	1	1
D	1	1	1	0

Bangladesh University
Department of CSE
Examination: Final
Semester: Fall 2024
Course code: 2106

Course title: Linear Algebra, Vector, Complex Variable and Laplace transformation

Time: 2 hours

Full marks: 50

Note: Answer any 4 questions. Each questions has 12.5 marks

1. a. Prove that $\sin^3 \theta = \frac{3}{4} \sin \theta - \frac{1}{4} \sin 3\theta$

b. Solve the equation $z^2 + (2i - 3)z + 5 - i = 0$ where $z \in \mathbb{C}$ 6+6.5 = 12.5

2. Prove that in any vector space.

a. The zero vector is unique

b. The inverse vector of an element is unique 6+6.5 = 12.5

3. What is Echelon matrix? Explain it with example.

Find the rank of the following matrix $A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ 12.5

4. Solve the following equations with the help of inverse matrix 12.5

$$3x + 5y - 7z = 13$$

$$4x + y - 12z = 6$$

$$2x + 9y - 3z = 20$$

5. If $s > 0$ then show that $L\{\sin at\} = \frac{a}{s^2 + a^2}$ (By the formula) 12.5

6. a. If $L\{f(t)\} = f(s)$ then show that $L\{F(at)\} = \frac{1}{a} f\left(\frac{s}{a}\right)$ 6+6.5 = 12.5

b. If $L\{F(t)\} = f(s)$ then show that, $L\{F'''(t)\} = s^2 F(s) - sF(0) - F'(0)$